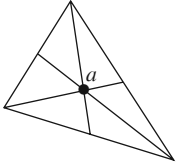
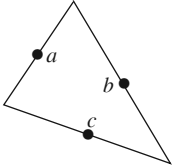
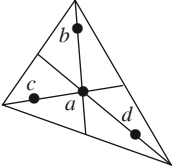
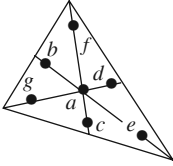


Table 9.2 Numerical integration formulae for triangles

Order	Figure	Error	Points	Triangular coordinates	Weights
Linear		$R = O(h^2)$	a	$\frac{1}{3}, \frac{1}{3}, \frac{1}{3}$	1
Quadratic		$R = O(h^3)$	a b c	$\frac{1}{2}, \frac{1}{2}, 0$ $0, \frac{1}{2}, \frac{1}{2}$ $\frac{1}{2}, 0, \frac{1}{2}$	$\frac{1}{3}$ $\frac{1}{3}$ $\frac{1}{3}$
Cubic		$R = O(h^4)$	a b c d	$\frac{1}{3}, \frac{1}{3}, \frac{1}{3}$ $0.6, 0.2, 0.2$ $0.2, 0.6, 0.2$ $0.2, 0.2, 0.6$	$-\frac{27}{48}$ $\frac{25}{48}$
Quintic		$R = O(h^6)$	a b c d e f g	$\frac{1}{3}, \frac{1}{3}, \frac{1}{3}$ $\alpha_1, \beta_1, \beta_1$ $\beta_1, \alpha_1, \beta_1$ $\beta_1, \beta_1, \alpha_1$ $\alpha_2, \beta_2, \beta_2$ $\beta_2, \alpha_2, \beta_2$ $\beta_2, \beta_2, \alpha_2$	0.225 000 000 0 0.132 394 152 7 0.125 939 180 5
with $\alpha_1 = 0.059\,715\,871\,7$ $\beta_1 = 0.470\,142\,064\,1$ $\alpha_2 = 0.797\,426\,985\,3$ $\beta_2 = 0.101\,286\,507\,3$					

Once again we could use n Gauss points and arrive at a summation expression of the type used in the previous section. However, the limits of integration now involve the variable itself and it is convenient to use alternative sampling points for the second integration by use of a special Gauss expression for integrals of the type given by Eq. (9.37) in which w is a linear function. These have been devised by Radau¹³ and used successfully in the finite element context.¹⁴ It is, however, much more desirable (and aesthetically pleasing) to use special formulae in which no bias is given to any of the natural coordinates L_i . Such formulae were first derived by Hammer *et al.*¹⁵ and Felippa¹⁶ and a series of necessary sampling points and weights is given in Table 9.2.¹⁷ (A more comprehensive list of higher formulae derived by Cowper is given on p. 184 of reference 17.)

A similar extension for tetrahedra can obviously be made. Table 9.3 presents some formulae based on reference 15.